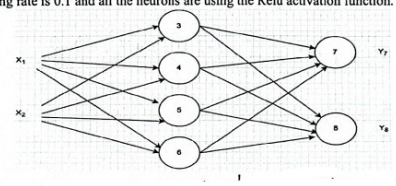
Solution

Brac University
CSE427: Machine Learning
Spring 2026
Quiz 02
Set B

Duration: 25 minutes

Name:	ID:	Section: 05

Question 1:
In the following neural network **find** the updated weights of W_{23} **and** W_{37}. Initially the weights of all the links are 1.5 and the biases of node 3, 4, 5 and 6 are -1 each. Assume the biases of node 7 and 8 are -3.0 and -2.5 respectively. Assume that the input is $X_1 = 1.5$ and $X_2 = 1.25$ and the target output is $Y_7 = 1$ and $Y_8 = 0$. The learning rate is 0.1 and all the neurons are using the Relu activation function.



all weight = 1·5

$b_3 = b_4 = b_5 = -1$

$b_7 = -3$

$b_8 = -2.5$

$Ob_7 = 1$

$Ob_8 = 0$

We know,

$$W_{23} = W_{23} + \eta \delta_3 x_2$$

$$W_{37} = W_{37} + \eta \delta_7 Y_3$$

$$Y_3 = ReLU(w_{13} x_1 + w_{23} x_2 + b_3) = ReLU(1.5 \times 1.5 + 1.5 \times 1.25 - 3)$$
$$= ReLU(3.125) = 3.125$$

$Y_3 = Y_4 = Y_5 = Y_6 = 3.125$ [as all weights are same]

$$Y_7 = ReLU(w_{37} Y_3 + w_{47} Y_4 + b_7) = ReLU(1.5 \times 3.125 + 1.5 \times 3.125 + (-3))$$
$$= ReLU(15.75) = 15.75$$

$$Y_8 = ReLU(w_{38} Y_3 + w_{48} Y_4 + b_8) = ReLU(1.5 \times 3.125 + 1.5 \times 3.125 + (-2.5))$$

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$$= ReLU(16.25) = 16.25$$

$$\delta_7 = Err_7 Y_7' = (Ob_7 - Y_7) Y_7' = (1 - 15.75) \times 1 = -14.75$$

$$\delta_8 = Err_8 Y_8' = (Ob_8 - Y_8) Y_8' = (0 - 16.25) \times 1 = -16.25$$

[if $x > 0$ $ReLU'(x) = 1$
$x < 0$ $ReLU'(x) = 0$]

$$\delta_3 = (\delta_7 w_{37} + \delta_8 w_{38}) Y_3'$$
$$= (-14.75) \times 1.5 + (-16.25) \times 1.5) \times 1$$
$$= -46.5$$

So,

$$W_{23} = W_{23} + \eta \delta_3 x_2$$
$$= 1.5 + 0.1 \times (-46.5) \times 1.25$$
$$= -4.3125$$

$$W_{37} = W_{37} + \eta \delta_7 Y_3$$
$$= 1.5 + 0.1 \times (-14.75) \times 3.125$$
$$= -3.109375$$

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